

Program 6:

Working with Gradle: Setting Up a Gradle Project, Understanding Build Scripts (Groovy and Kotlin DSL), Dependency Management and Task Automation

Step 1: create directory using command **mkdir MyGradleApp** and enter the directory using the command **cd MyGradleApp**

```
samiksha@samiksha:~/Devops6D$ mkdir MyGradleapp
samiksha@samiksha:~/Devops6D$ cd MyGradleapp
samiksha@samiksha:~/Devops6D/MyGradleapp$
```

Step 2: Type the command **gradle init --type java-application**

```
samiksha@samiksha:~/Devops6D/MyGradleapp$ gradle init --type java-application

Enter target Java version (min: 7, default: 21): 21

Project name (default: MyGradleapp): MyGradleapp

Select application structure:
  1: Single application project
  2: Application and library project
Enter selection (default: Single application project) [1..2] 1

Select build script DSL:
  1: Kotlin
  2: Groovy
Enter selection (default: Kotlin) [1..2] 2

Select test framework:
  1: JUnit 4
  2: TestNG
  3: Spock
  4: JUnit Jupiter
Enter selection (default: JUnit Jupiter) [1..4] 1

Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no] no

> Task :init
To learn more about Gradle by exploring our Samples at https://docs.gradle.org/8.7/samples/sample_building_java_applications.html

BUILD SUCCESSFUL in 2m 9s
1 actionable task: 1 executed
<-----> 0% WAITING
> IDLE
samiksha@samiksha:~/Devops6D/MyGradleapp$
```

Step 3: Go to app directory to find build.gradle. Open build.gradle and type the code in build.gradle

```
task hello {
    doLast {
        println '6D Devops Class'
    }
}
```

Step 4: type the command **gradle build** to compile the source code and package the app.

```
samiksha@samiksha:~/Devops6D/MyGradleapp$ gradle build
Starting a Gradle Daemon, 1 incompatible and 1 stopped Daemons could not be reused, use --status for details
Path for java installation '/usr/lib/jvm/openjdk-11' (Common Linux Locations) does not contain a java executable
Path for java installation '/usr/lib/jvm/openjdk-17' (Common Linux Locations) does not contain a java executable

BUILD SUCCESSFUL in 1m 5s
7 actionable tasks: 7 executed
samiksha@samiksha:~/Devops6D/MyGradleapp$
```

Step 5: type the command **gradle run** to execute the default Hello World application

```
samiksha@samiksha:~/Devops6D/MyGradleapp$ gradle run
Path for java installation '/usr/lib/jvm/openjdk-17' (Common Linux Locations) does not contain a java executable
Path for java installation '/usr/lib/jvm/openjdk-11' (Common Linux Locations) does not contain a java executable

> Task :app:run
Hello World!

BUILD SUCCESSFUL in 1s
2 actionable tasks: 1 executed, 1 up-to-date
```

Step 6: type the command **gradle <task-name>** to run the task

```
samiksha@samiksha:~/Devops6D/MyGradleapp$ gradle hello
```

```
> Task :app:hello
6D Devops Class
```

```
BUILD SUCCESSFUL in 1s
1 actionable task: 1 executed
```

Step 7: in the terminal type the code MyGradleApp to enter the folder .

Type the command `git clone JenkinsFile` to open the Jenkinsfile and paste the code

pipeline {

agent any // Use any available agent

tools {

gradle 'Gradle' // Ensure this matches the name configured in Jenkins

jdk 'JDK'

}

stages {

stage('Checkout') {

steps {

git branch: 'master', url: 'https://github.com/Samiksha201104 /MyGradleApp.git'

}

}

stage('Build') {

steps {

sh 'gradle build' // Run Gradle build

}

}

stage('Test') {

```
    steps {
        sh 'gradle test' // Run unit tests
    }
}

stage('Run Application') {
    steps {
        // Start the JAR application
        sh 'gradle run'
    }
}

post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}
```

Commit and push to GitHub

Step 8: in Jenkins Dashboard click on New Item

Give the name MyGradleApp and select Pipeline



Jenkins / All / New Item

New Item

Enter an item name

MyGradleApp

Select an item type

**Pipeline**

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

**Freestyle project**

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

**Maven project**

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

**Multi-configuration project**

OK

In configure scroll down to Pipeline and choose pipeline script from SCM, give the repository url and choose GitHub Credentials



Jenkins / MyGradleApp / Configure



Configure

General

Triggers

Pipeline

Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

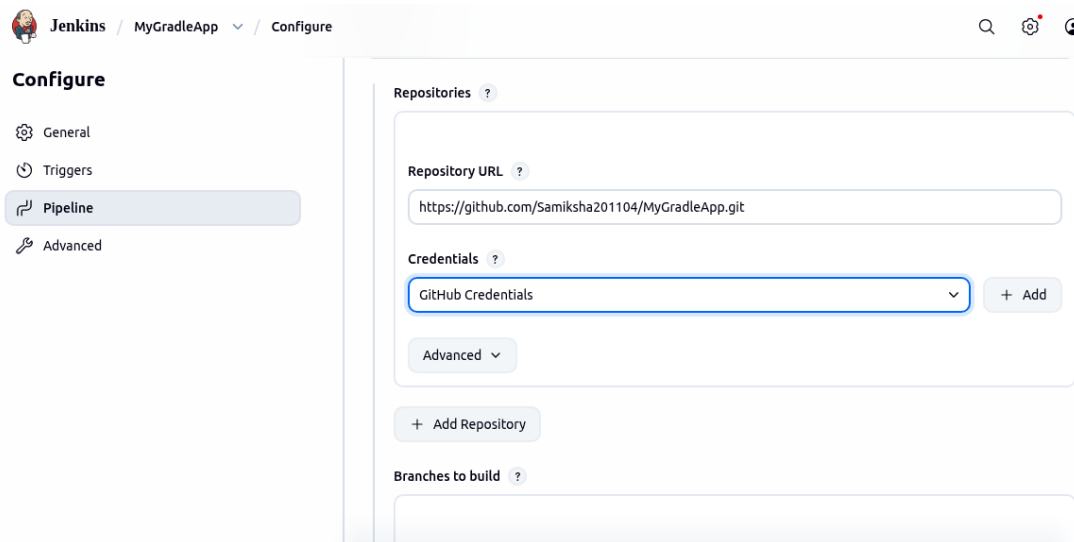
Script Path ?

Jenkinsfile

☒ Lightweight checkout ?[Pipeline Syntax](#)**Advanced**

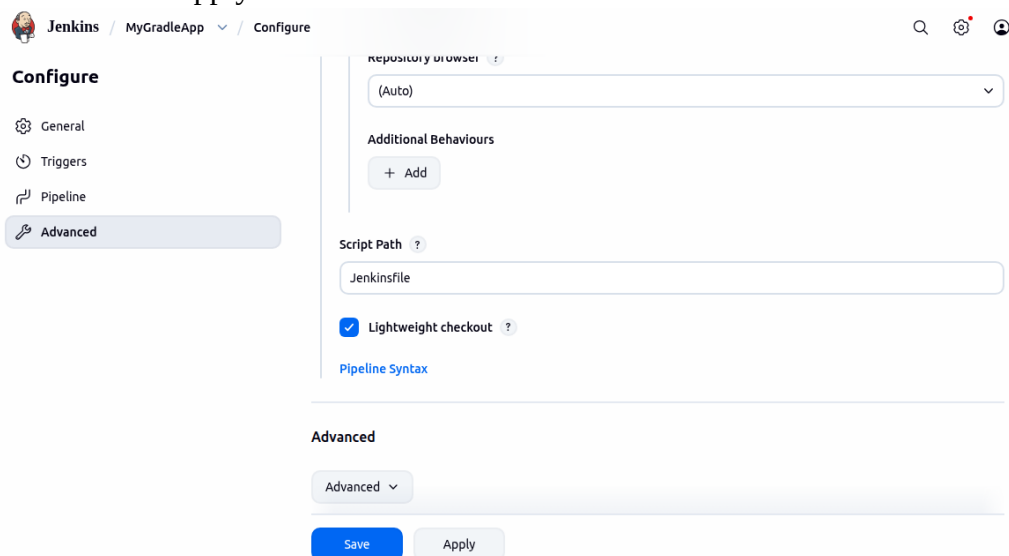
Save

Apply



The screenshot shows the Jenkins configuration page for a pipeline named 'MyGradleApp'. The left sidebar has tabs for 'General', 'Triggers', 'Pipeline', and 'Advanced'. The 'Pipeline' tab is selected. The main content area is titled 'Configure' and shows the 'Repositories' section. A 'Repository URL' field contains 'https://github.com/Samiksha201104/MyGradleApp.git'. A 'Credentials' dropdown menu is set to 'GitHub Credentials'. There is an 'Add' button next to the credentials dropdown. Below this is an 'Advanced' dropdown menu. At the bottom of the 'Repositories' section is a '+ Add Repository' button. The 'Branches to build' section is partially visible at the bottom.

Then click on apply and save



The screenshot shows the Jenkins configuration page for the same pipeline, now with the 'Advanced' tab selected. The 'Repository browser' dropdown is set to '(Auto)'. There is an 'Additional Behaviours' section with an 'Add' button. The 'Script Path' field contains 'Jenkinsfile'. The 'Lightweight checkout' checkbox is checked. Below this is a 'Pipeline Syntax' link. At the bottom, there is an 'Advanced' dropdown menu and two buttons: 'Save' and 'Apply'.

Step 9: go to Jenkins Dashboard select the MyGradleApp pipeline and select Build Now. After build select Full Stage View

Jenkins

+ New Item

Build History

Project Relationship

Check File Fingerprint

Build Queue: No builds in the queue.

Build Executor Status: 0/2

All +

S	W	Name	Last Success	Last Failure	Last Duration
...	☀	MyGradleApp	N/A	N/A	N/A
✓	☁	MyMavenApp	23 hr	#1	33 sec

Icon: S M L

- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Stages
- Rename
- Pipeline Syntax

Once the errors are corrected Build again

Jenkins / MyGradleApp / Full Stage View

MyGradleApp - Stage View

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Run Application	Declarative: Post Actions
Average stage times:	3s	536ms	2s	2min 50s	51s	15s	7s
#4 Apr 10 21:49 1 commit	3s	314ms	2s	1min 40s	1min 14s	50s	13s
#3 Apr 10 21:29 2 commits	2s	301ms	2s	4min 48s	1min 36s failed	8s failed	8s
#2 Apr 10 21:10 2 commits	4s	743ms	1s	4min 43s failed	32s failed	2s failed	6s
#1 Apr 10 11:28 No Changes	2s	786ms	2s	10s failed	654ms failed	204ms failed	522ms

